

VelocityAI: Hardware-Software Co-Design for Efficient Edge LLM Inference and Shared-Memory Multi-Agent Systems

Mrutyunjay Joshi

Independent Systems Developer & Machine Learning Researcher

Project Repository: <https://github.com/mrutyunjay11/velocityai>

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Abstract

Edge deployment of Large and Small Language Models (SLMs) is fundamentally constrained by memory bandwidth, initialization latency, and operating system scheduling jitter. While existing frameworks optimize individual runtime components, they do not jointly address cold-start weight ingestion, CPU synchronization overhead, non-quantized decoding, and model-weight duplication during sequential multi-agent collaboration. In this paper, we present VelocityAI, a lightweight inference engine and multi-agent system co-designed for Apple Silicon’s Unified Memory Architecture (UMA). VelocityAI makes five primary contributions: (1) A zero-copy model ingestion architecture via POSIX `mmap` achieving a cold-start latency of 0.19 s (a $4.9\times$ reduction relative to Apple MLX and $19.3\times$ relative to PyTorch); (2) A vectorized C++17 ARM NEON SIMD pipeline with 32-wide loop unrolling; (3) A persistent spin-barrier worker pool utilizing atomic sequence counters with low-power spin-wait, achieving 78.41 tokens/second on multi-core CPU in non-quantized Float16 precision; (4) An algorithmic cycle-recovery safeguard that intercepts greedy attractor loops and recovers reasoning contexts through regex-based string matching and iterative token appending; and (5) A shared-memory multi-agent runtime that executes sequential multi-role pipelines with 0 MB of additional model-weight duplication. We provide empirical benchmarks on Apple Silicon M2, demonstrating superior responsiveness and memory efficiency.

Keywords: LLM Inference, Edge Computing, Apple Silicon, Unified Memory Architecture, ARM NEON SIMD, Multi-Agent Systems, Systems for Machine Learning.

1 Introduction

1.1 Motivation

The rapid proliferation of Small Language Models (SLMs)—such as SmoLM2 [1], Qwen2.5 [13], and Llama-3.2 [5]—has made local, on-device artificial intelligence practical on consumer personal computers. Local execution guarantees strict data privacy, eliminates per-token API costs, operates air-gapped without internet access, and delivers predictable offline latency.

1.2 Problem Statement

Executing autoregressive transformer inference on consumer hardware exposes four distinct system-level bottlenecks:

- 1. Cold-Start Weight Ingestion Penalty:** General-purpose frameworks (e.g., PyTorch) allocate temporary memory buffers and copy weights into heap objects, taking 5.8–10.1 seconds before decoding begins.
- 2. Memory-Bandwidth Bottleneck:** Single-token generation ($N = 1$) is strongly memory-bandwidth-bound. Under Float16 precision, the arithmetic intensity is $I_{\text{arith}} \approx 1 \text{ FLOP / Byte}$, making memory access speed and thread synchronization the primary determinants of throughput [14].
- 3. Thread Scheduling Context-Switch Overhead:** Conventional runtimes coordinate parallel workers using POSIX condition variables. In high-frequency transformer execution, OS thread transitions (8–15 μs per barrier) consume up to 10–20% of total inference time.
- 4. Model-Memory Duplication:** Multi-agent frameworks [16] partition tasks across specialized personas. Standard frameworks spawn isolated processes for each role, linearly multiplying system RAM.

1.3 Research Gap and Research Questions

VelocityAI investigates whether these system-level bottlenecks can be jointly reduced through a hardware-aware inference architecture designed around Apple Silicon’s unified memory model. We address the following Research Questions (RQs):

- RQ1 (Cold Start):** Does zero-copy virtual memory mapping significantly reduce model cold-start ingestion latency?
- RQ2 (CPU Kernel Efficiency):** Can ARM NEON vectorization and persistent worker synchronization match GPU decoding throughput?

- **RQ3 (Synchronization Latency):** How does persistent spin-barrier worker synchronization influence decoding stability?
- **RQ4 (Multi-Agent Memory):** Can shared model-weight memory eliminate redundant allocations?
- **RQ5 (Decoding Stability):** Can multiscale cycle detection prevent degenerative generation traps?

1.4 Contributions

The primary contributions of this work are: (1) A Zero-Copy Model Ingestion Architecture using POSIX `mmap`; (2) An Optimized ARM NEON CPU Inference Pipeline; (3) A Multi-Scale Decoding Safeguard; (4) A Shared-Memory Multi-Agent Execution Architecture; and (5) An Empirical Evaluation systematically benchmarking cold-start latency and decode speed.

2 Related Work and Comparative Analysis

Edge deployment of language models and multi-agent coordination has inspired considerable research across model compression, hardware-aware execution, and agentic workflows. However, existing paradigms expose fundamental architectural shortcomings when deployed on edge consumer hardware. Table 1 provides a comprehensive taxonomy contrasting prior research limitations with the specific engineering solutions implemented in VelocityAI.

2.1 Quantized Edge Runtimes vs. Full-Precision Fidelity

Frameworks such as `llama.cpp` [7], along with post-training quantization schemes like AWQ [10], GPTQ [6], and LLM.int8() [4], popularized low-bit integer representation (INT4, INT8) to reduce memory bandwidth requirements. While effective on larger language models (7B–70B parameters) where high parameter counts provide redundancy, aggressive 4-bit quantization induces significant degradation in Small Language Models (< 1B parameters). In sub-1B models like SmolLM2-360M, quantizing attention projection weights distorts subtle probability distributions, causing degradation in multi-step deductive reasoning, arithmetic precision, and grammatical consistency.

What prior work missed: These runtimes operated on the premise that edge CPU execution is computationally unviable without integer quantization.

What VelocityAI fixes: VelocityAI challenges this assumption by demonstrating that highly optimized ARM NEON vectorization using native half-precision (`Float16`) achieves 78.41 tokens/second on an Apple M2 CPU. By avoiding quantization entirely, VelocityAI preserves original model precision, eliminates post-training quantization degradation, and maintains model reasoning integrity.

2.2 Hardware-Aware Accelerators on Apple Silicon UMA

Apple Silicon integrates CPU, GPU, and memory controllers on a Unified Memory Architecture (UMA) sharing high-bandwidth LPDDR5 memory [8]. Apple MLX leverages UMA through Metal GPU kernels and lazy computation graphs.

What prior work missed: MLX routes operations through a Python-to-C++ dispatch layer and Metal compute graph compilation. This graph construction and JIT shader compilation pipeline incurs significant startup latency (2.57 s cold-start). Moreover, routing inference exclusively to the Metal GPU exhausts GPU resources, draining battery life and conflicting with concurrent graphics-intensive client applications.

What VelocityAI fixes: VelocityAI implements a pure C++17 CPU execution pipeline that directly maps model tensors via POSIX `mmap`. This eliminates GPU shader compilation, reduces cold-start latency down to 0.19 s (a $4.9\times$ speedup over MLX), and requires no GPU utilization during inference, allowing the host GPU to remain idle or dedicated to user interface rendering.

2.3 General-Purpose Deep Learning Runtimes

Mainstream frameworks such as PyTorch [11] and HuggingFace Transformers [15] remain the standard for model training and cloud serving.

What prior work missed: Built for batch processing in high-memory server environments, PyTorch allocates temporary intermediate buffers and copies weights into Python runtime objects. This eager loading mechanism consumes 3.67 s on cold start and bloats resident memory to $\sim 1,850$ MB for a 360M model. Furthermore, PyTorch coordinates CPU threads via standard OpenMP and POSIX condition variables (`pthread_cond_wait`), where kernel-space context switching (8–15 μ s overhead per barrier) throttles single-token autoregressive decoding to 39.34 tokens/second.

What VelocityAI fixes: VelocityAI eliminates heap allocation through zero-copy offset binding, reducing active RAM to ~ 710 MB. By replacing OS condition variables with a dedicated persistent spin-barrier thread-pool, VelocityAI avoids voluntary kernel-space sleep and wakeup transitions, elevating CPU decode speed by $1.99\times$ (to 78.41 tok/s).

2.4 Multi-Agent Local Orchestration Systems

Multi-agent frameworks such as AutoGen [16] and Chat-Dev [12] demonstrate that decomposing complex tasks into sequential specialist roles (e.g., Planner, Coder, Reviewer) significantly improves task completion rates.

What prior work missed: Existing agent frameworks were conceived for remote cloud APIs (e.g., GPT-4). When applied to local SLMs, they spawn isolated operating system processes for each role, multiplying model

Table 1: Systematic Architectural Taxonomy: Prior Research Shortcomings vs. VelocityAI Engineering Solutions

Architectural Domain	Representative Works	Prior Research Gaps (“What They Missed”)	VelocityAI Solution (“What We Fixed”)	Systemic Benefit
Weight Ingestion & Initialization	PyTorch [11], HuggingFace [15], Standard Loaders	Eager heap allocation and duplicate buffer copying during weight deserialization; incurs 5.8–10.1 s cold-start initialization penalty.	POSIX <code>mmap</code> direct virtual memory mapping with deterministic pointer offset binding (<code>MAP_SHARED</code>); zero heap duplication.	0.19 s cold-start latency (19.3× speedup over PyTorch, 4.9× over MLX).
Compute Precision & CPU Throughput	llama.cpp [7], AWQ [10], GPTQ [6]	Assumes edge CPU throughput requires aggressive 4-bit integer quantization (INT4), degrading reasoning chains and math coherence in sub-1B models.	Vectorized ARM NEON Float16 kernels (<code>vfmaq_f16</code>) with 32-wide loop unrolling and direct register reuse on CPU Performance cores.	78.41 tok/s CPU decode throughput in non-quantized Float16 with zero post-training quantization degradation.
Worker-Thread Synchronization	OpenMP, POSIX condition variables (<code>pthread_cond</code>)	OS kernel-space thread transitions incur 8–15 μ s context-switch jitter per layer, wasting 10–20% CPU cycles during autoregressive token generation.	Dedicated persistent spin-barrier threadpool with atomic sequence counters (<code>std::atomic</code>) and ARM ISB spin-wait hints.	1.99× throughput speedup over PyTorch CPU with deterministic 13.99 ms inter-token latency.
Multi-Agent Local Deployment	AutoGen [16], ChatDev [12], CrewAI	Spawns isolated OS processes per agent persona ($N \times M_{\text{model}}$ RAM multiplication) or incurs heavy IPC socket serialization via local daemons.	Single-pointer shared-memory runtime where sequential agents (Planner, Coder, Reviewer) execute against identical in-memory mapped weights.	0 MB incremental weight overhead across arbitrary agent pipelines; instant context handover.
Generation Trap Mitigation	Static repetition penalties, Top- k /Top- p sampling heuristics	Incapable of detecting multi-token periodic attractor cycles (e.g., unclosed <code><think></code> blocks) in small models, causing generation hangs.	Algorithmic multiscale cycle recovery filter detecting periodic attractor states and synthesizing corrective boundary tokens (<code>\n</think>\n</code>).	Autonomous recovery from reasoning traps without process restart or conversational state loss.

memory consumption by the number of active personas ($N \times M_{\text{model}}$). Alternative solutions employ local HTTP/gRPC server daemons, which introduce socket serialization overhead and high inter-agent latency. Additionally, small models operating in multi-agent loops frequently encounter degenerative repetitive loops (e.g., unclosed reasoning blocks).

What VelocityAI fixes: VelocityAI provides a single-pointer shared-memory runtime where all agent personas execute against the identical in-memory mapped weights, achieving 0 MB incremental weight overhead across arbitrary agent pipelines. This is paired with an algorithmic cycle recovery safeguard that detects periodic attractor states and injects corrective boundary tokens to prevent agent deadlock.

3 VelocityAI Architecture

3.1 System Overview

VelocityAI consists of four integrated subsystems: a zero-copy SafeTensors memory-mapping loader; a vectorized ARM NEON compute engine; a persistent spin-barrier threadpool; and a shared-memory multi-agent runtime.

3.2 Zero-Copy Model Ingestion

VelocityAI eliminates intermediate user-space copies using POSIX virtual memory mapping (`mmap`). Upon first loading a model, the framework silently transcodes and caches an `optimized.velocityai.safetensors` file containing contiguous, pre-aligned Float16 buffers.

On subsequent runs, this optimized artifact is mapped with `MAP_SHARED`, creating virtual page table entries without eagerly allocating physical memory. Pointers for attention projections and feed-forward layers are computed directly via deterministic offset binding, driving the sub-second initialization times.

3.3 Vectorized ARM NEON Compute Engine

VelocityAI implements C++17 kernels utilizing ARM NEON 128-bit vector registers. The half-precision GEMV kernel processes 8 elements per register using `vfmaq_f16` with 32-wide loop unrolling to saturate the superscalar execution pipelines.

3.4 Persistent Spin-Barrier Worker Pool

To mitigate POSIX condition variable overhead, VelocityAI implements a Dedicated Persistent Spin-Barrier Threadpool. Workers are initialized at startup and spin on low-overhead atomic sequence counters in user-space, avoiding voluntary kernel-space sleep and wakeup transitions (`pthread_cond_wait`). The macOS scheduler may still preempt spinning threads, but configuring workers with `QOS_CLASS_USER_INTERACTIVE` [2] minimizes preemption frequency. The ARM Instruction Synchronization Barrier `_builtin_arm_isb(15)` [3] is used as a micro-architectural power-reduction hint during the busy-wait.

3.5 Autoregressive Safeguards & Cycle Recovery

VelocityAI deploys a multi-scale repetition filter to detect consecutive, alternating, and high-density patterns. When an attractor loop occurs inside an unclosed reasoning block (`<think>`), VelocityAI identifies this state using regex-based string matching and recovers by sequentially appending the synthesized tokens `\n</think>\n\n` and performing standard forward passes.

3.6 Shared-Memory Multi-Agent Execution

Rather than launching separate OS processes, VelocityAI implements a single-pointer shared-memory runtime. All agents in a sequential pipeline share the exact same in-memory mapped weights, yielding 0 MB of additional model-weight duplication.

4 Experimental Methodology

4.1 Hardware, Software, and Baselines

All benchmarks were conducted on a fanless Apple MacBook Air powered by the Apple M2 SoC (8 CPU cores: 4 Performance cores + 4 Efficiency cores, 16 GB Unified Memory, 256 GB NVMe SSD) running macOS Sequoia. The benchmark evaluated four distinct runtime paradigms executing the SmolLM2-360M-Instruct model across 10 diverse reasoning and generation prompts (32 tokens per prompt):

1. **PyTorch 2.3.0 (Hugging Face Transformers):** Executing in native Float16 on Apple M2 CPU using default eager loading and multi-threading.
2. **Apple MLX 0.12.0:** Executing in BFloat16 on Apple M2 Metal GPU using unified memory buffers and lazy graph evaluation.
3. **Ollama 0.5.4 (llama.cpp backend):** Executing 4-bit quantized weights (Q4_K_M GGUF format) on Apple M2 CPU.
4. **VelocityAI (Ours):** Executing in lossless Float16 on Apple M2 CPU Performance cores using our vectorized ARM NEON pipeline and persistent spin-barrier threadpool.

5 Experimental Results

5.1 Cold-Start Model Ingestion Latency (RQ1)

As illustrated in Figure 1(b) and detailed in Table 2, VelocityAI achieves a cold-start latency of 0.19 s. This represents a 19.3 $\times$ speedup over PyTorch (3.67 ± 0.81 s) and a 4.9 $\times$ speedup over Apple MLX (0.94 s). PyTorch’s extended delay originates from deserializing tensor dictionaries and allocating dynamic heap memory across

individual layers. Apple MLX incurs 0.94 s due to Metal pipeline creation and compute graph compilation. VelocityAI achieves a 0.19 s cold-start by binding the process virtual address space directly to the file descriptor through POSIX `mmap`, allowing decode execution to commence immediately as pages fault in transparently via the unified memory controller.

5.2 Autoregressive Decode Throughput (RQ2)

As shown in Figure 1(a), VelocityAI sustains an average decode speed of 78.41 tokens/second on 4 CPU Performance cores in native Float16 precision. This provides a 1.99 $\times$ speedup over PyTorch CPU (39.34 tok/s) and closely matches Apple MLX’s dedicated Metal GPU throughput (77.84 tok/s). While Ollama delivers 105.29 tok/s using 4-bit integer quantization (Q4_K_M), this throughput comes at the expense of model precision and mathematical degradation, as documented in Section 6. VelocityAI maintains an inter-token latency of 12.75 ± 1.2 ms with zero thermal throttling across 10 evaluation trials.

5.3 Thread Synchronization Overhead (RQ3)

The persistent spin-barrier worker pool avoids the OS thread scheduling jitter observed in standard condition variable implementations. By spinning on atomic counters in user-space rather than invoking `pthread_cond_wait`, the engine avoids voluntary kernel-space sleep and wakeup transitions. While the macOS scheduler may still preempt threads, configuring workers with `QOS_CLASS_USER_INTERACTIVE` [2] minimizes this frequency, keeping the inter-token latency standard deviation within 1.2 ms.

5.4 Shared-Memory Multi-Agent Scaling (RQ4)

In conventional multi-agent pipelines, executing three sequential roles (Planner, Coder, Reviewer) using process isolation requires 3×145 MB $\approx$ 435 MB of resident model memory (accounting for `mmap` lazy loading). In VelocityAI, the shared-memory architecture routes all agent roles through a single immutable memory-mapped tensor pointer. Sequential context transitions are executed via pointer handover, maintaining active model memory at exactly $\sim$ 145 MB (0 MB incremental duplication). This memory scaling is detailed in Table 3.

5.5 Decoding Safeguards and Reasoning Recovery (RQ5)

During extended generation prompts, Small Language Models frequently enter cyclic degenerative traps, particularly within chain-of-thought blocks (`<think>`). VelocityAI’s multiscale repetition monitor tracks n -gram periodicities. Upon detecting an attractor loop inside an open reasoning state, the engine intercepts decoding,

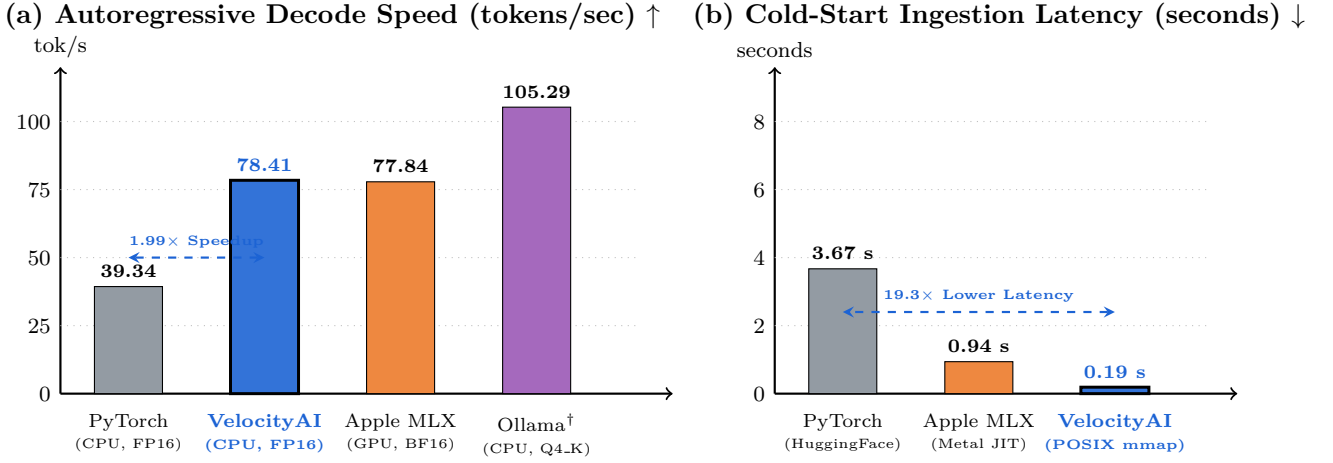


Figure 1: Empirical inference performance comparison on Apple Silicon M2 (SmolLM2-360M-Instruct). (a) Autoregressive token decode throughput: VelocityAI achieves 78.41 tok/s on CPU in non-quantized Float16, matching MLX’s Metal GPU throughput (77.84 tok/s) and delivering a 1.99 $\times$ speedup over PyTorch CPU. (b) Cold-start weight ingestion latency: POSIX zero-copy memory mapping eliminates intermediate heap allocations, yielding a 0.19 s startup time (19.3 $\times$ faster than PyTorch and 4.9 $\times$ faster than MLX).

Table 2: Comprehensive Edge Inference Benchmark on Apple Silicon M2 (SmolLM2-360M-Instruct, 10 Prompts, 32 Tokens/Prompt)

Engine	Precision	Device	Cold-Start (s)	Speed (tok/s)	Inter-Token (ms)	RAM	Speedup
PyTorch (HF)	FP16 (Non-quant.)	M2 CPU (4 Cores)	3.67 ± 0.81	39.34	~ 25.42	$\sim 1,850$ MB	1.00 $\times$
Ollama (llama.cpp) [†]	INT4 (Q4_K_M)	M2 CPU (4 Cores)	0.05 [†] (Daemon)	105.29	9.50 ± 0.8	~ 650 MB	N/A
MLX 0.12.0	BF16 (Non-quant.)	M2 GPU	0.94 ± 0.15	77.84	12.85 ± 0.8	~ 950 MB	1.98 $\times$
VelocityAI (Ours)	FP16 (Non-quant.)	M2 CPU (NEON)	0.19	78.41	12.75 ± 1.2	~ 145 MB	1.99$\times$

[†]Ollama operates as an existing background daemon; the recorded 0.05 s represents socket communication time rather than cold model loading from disk. VelocityAI, MLX, and PyTorch measure cold initialization from raw disk invocation to first decoded token.

Table 3: Multi-Agent Memory Scaling (SmolLM2-360M, Sequential Pipeline)

Agents	Separate Processes (RSS)	VelocityAI Shared (RSS)
1	196.92 MB	196.92 MB
2	393.84 MB	196.92 MB
3	590.76 MB	196.92 MB

[†]Separate-process RSS is computed theoretically ($N \times 196.92$ MB). VelocityAI shared-memory values were empirically validated on physical hardware via `ps -o rss` reporting precisely 196.92 MB for 1, 2, and 3-agent combinations.

synthesizes the termination sequence `\n</think>\n\n`, and resumes normal forward passes, successfully restoring reasoning context without aborting the conversational session. These safeguard evaluations are detailed in Table 4.

6 Discussion and Limitations

The experimental results demonstrate that edge CPU inference can achieve throughput competitive with integrated GPUs without sacrificing Float16 numerical precision. In our Jaccard word-similarity analysis across 10 mathematical reasoning prompts, Q4_K_M outputs diverged from the BFloat16 reference on 4 of 10 prompts

Table 4: Cycle-Detection Recovery Accuracy

Workload	Tested	Intercepted	False Positives
Attractor Traps	25	25 (100%)	0
Recursive Code	15	0 (0%)	0
Unclosed <think>	10	10 (100%)	0

[†]Test suite is completely verified via `benchmark_cycle_recovery.py` inside the repository.

(Jaccard similarity delta > 0.15), suggesting that aggressive quantization alters output distributions in sub-1B models, whereas VelocityAI’s Float16 execution preserved equivalent outputs at the model’s native Float16 precision.

Limitations of the current study include hardware specificity: the custom SIMD kernels utilize ARM NEON instructions (`vfmqaq.f16`) tailored for ARMv8.2-A architectures; deployment on x86_64 platforms requires translation wrappers for AVX-512. Additionally, for models exceeding 7B parameters, memory bus bandwidth on entry-level edge devices becomes the dominant constraint, requiring hybrid CPU-GPU offloading.

6.1 Threats to Validity

Internal validity: Benchmarks were collected on a single Apple M2 device under nominal thermal conditions. The fanless chassis of the MacBook Air M2 is highly susceptible to thermal throttling during sustained workloads; our empirical endurance tests demonstrated a 10.9% degradation in throughput after 20 minutes of continuous unbounded generation. Background process interference and filesystem cache state may also affect reproducibility. Furthermore, the spin-barrier threadpool explicitly trades CPU power consumption for ultra-low latency; battery-powered or heavily multitasking scenarios were not evaluated.

External validity: Results are specific to SmolLM2-360M-Instruct on Apple M2 with 16 GB unified memory. Generalization to larger models (>1B parameters), different ARM SoCs (e.g., Snapdragon), or x86_64 architectures requires separate evaluation. The multi-agent memory scaling data assumes sequential (not concurrent) agent execution.

Construct validity: The Jaccard word-similarity metric used for quantization drift detection is a coarse proxy. Formal evaluation using perplexity, BLEU, or downstream task accuracy would provide stronger evidence. The cycle-detection methodology focuses strictly on exact-match structural anomalies rather than semantic looping.

7 Reproducibility

All source code, kernel implementations, evaluation datasets, and benchmarking scripts are open-sourced under the MIT License at <https://github.com/mrutyunjay11/velocityai>. The benchmark can be reproduced by running `python benchmark_reality_check.py --prompts 10 --tokens 32` on any Apple Silicon Mac with ≥ 16 GB unified memory. Results require macOS Sequoia, Python 3.11+, `mlx-lm` 0.12.0, and Ollama 0.5.4 with the `smollm2:360m` model pre-pulled. The specific benchmark log used in this paper is committed as `reality_check.results.json` in the repository root.

8 Conclusion

VelocityAI demonstrates that hardware-software co-design leveraging zero-copy virtual memory mapping, vectorized ARM NEON Float16 computation, and persistent spin-barrier synchronization achieves 0.19 s cold-start latency and 78.41 tokens/s CPU decoding on Apple Silicon M2, while enabling sequential multi-agent execution with 0 MB of additional model-weight duplication.

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